MM: 720

TEST-2

Time: 3 Hrs.

[PHYSICS]

Choose the correct answer:

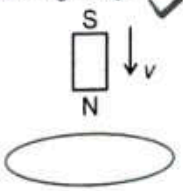
1. A square loop of side 4 m lies in the x-y plane in a region, where the magnetic field is given by, $\vec{B} = B_0(13\hat{i} - 7\hat{j} + 2\hat{k})$ T, where B_0 is constant. The magnitude of flux passing through the square loop is

(1) $(16B_0)$ Wb (2) $(96B_0)$ Wb

(3) $(208B_0)$ Wb (4) $(32B_0)$ Wb

2. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): When a magnet is made to fall freely through a closed conducting loop as shown in the figure its acceleration is always less than acceleration due to gravity.



Reason (R): Current is induced in the loop in such a way that it opposes the motion of the magnet, as per Lenz's law.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

3. Two identical circular loops of metal wire are lying on a horizontal table without touching each other. Loop-1 carries a current which increases with time. In response, the loop-2



(1) Remains stationary

(2) Is attracted by the loop-1

(3) Is repelled by loop-1

(4) Rotates about its centre of mass, with centre of mass fixed

4. A loop of area 4 m^2 is placed in a magnetic field, if the rate of change of flux in the loop is 20 Wb/s then magnitude of the induced emf in the loop is

(1) 20 V (2) 10 V

(3) 5 V (4) Zero

5. A uniform wire of resistance 100 Ω is melted and recast in a wire of length double that of original. The resistance of the wire after melting and recasting will be

(1) 400 Ω (2) 200 Ω

(3) 800 Ω (4) 100 Ω

6. The S.I. unit of magnetic induction (B) is (symbols have their usual meaning)

(1) $\frac{J}{A\,m^2}$

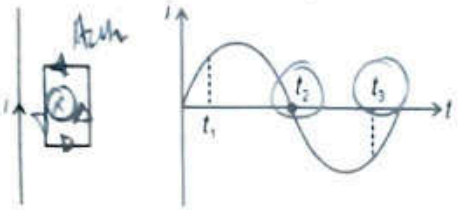
(2) $\frac{N}{A\,m^2}$

(3) $\frac{V\,s}{m}$

(4) $\frac{Wb}{m}$

Space for Rough Work

7. The current i in the straight conductor varies sinusoidally with time as shown. At time t_1 the induced current in the rectangular loop is anticlockwise. What is the current in the rectangular loop at time t_2 and t_3 respectively? (A positive value of i is in the direction indicated)

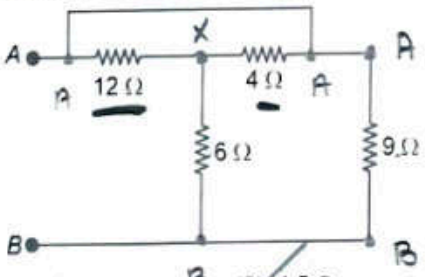


(1) Clockwise, Anticlockwise
(2) Clockwise, Clockwise
(3) Anticlockwise, Anticlockwise
(4) Anticlockwise, Clockwise

8. A small loop of wire of area $A = 0.02\ m^2$, number of turns, $N = 20$ and resistance, $R = 20\ \Omega$ is initially kept in a uniform magnetic field B in such a way that the field is normal to the plane of the loop. When it is pulled out of the magnetic field, a total charge of $q = 10^{-5}$ C flows through the loop. The magnitude of field B is

(1) 2×10^{-4} T
(2) 4×10^{-4} T
(3) 5×10^{-4} T
(4) Zero

9. Consider a part of electric circuit as shown. The equivalent resistance between points A and B is

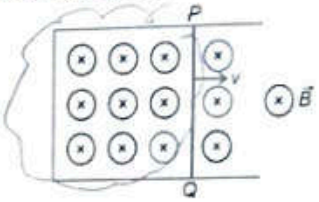


(1) 9 Ω
(2) 4.5 Ω
(3) 7 Ω
(4) 7.2 Ω

10. A galvanometer of resistance 25 Ω measures upto 20 mA. The value of shunt required in parallel to measure current upto 10 A is

(1) $\frac{25}{499}\Omega$
(2) $\frac{24}{499}\Omega$
(3) $\frac{7}{499}\Omega$
(4) $\frac{1}{20}\Omega$

11. A rod PQ lies across frictionless rails in a uniform magnetic field $\vec{B}$ as shown. The rod moves to the right with uniform speed v. In order for the induced emf in the circuit to be zero, the magnitude of the magnetic field should



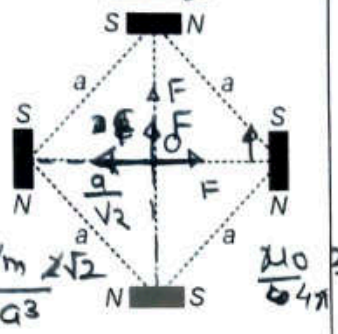
(1) Increase non-linearly with time
(2) Decrease non-linearly with time
(3) Increase linearly with time
(4) Decrease linearly with time

12. A rectangular loop of sides 2 m and 3 m is placed in xy plane. A very long wire is also placed in xy plane such that side of length 2 m of loop is parallel to the wire. The distance between the wire and the nearest side of loop is 5 m. The mutual inductance of this system is

(1) $\frac{\mu_0}{2\pi}\ln\left(\frac{8}{5}\right)$
(2) $\frac{\mu_0}{\pi}\ln\left(\frac{8}{5}\right)$
(3) $\frac{\mu_0}{2\pi}\ln\left(\frac{5}{8}\right)$
(4) $\frac{2\mu_0}{\pi}\ln\left(\frac{5}{8}\right)$

Space for Rough Work

13. Four small identical bar magnets each of magnetic dipole moment M are placed on vertices of a square of side a such that diagonals of the square coincide with perpendicular bisectors of respective magnets as shown below. The net magnetic field at the centre of the square is



(1) Zero

(2) $\frac{\mu_0 M}{\sqrt{2}\pi a^3}$

(3) $\frac{\mu_0 M}{2\pi a^3}$

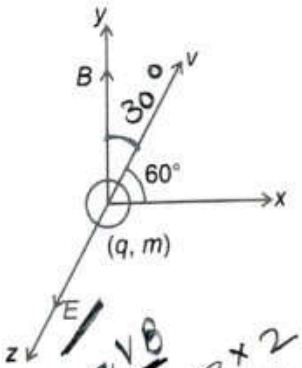
(4) $\frac{\sqrt{2}\mu_0 M}{\pi a^3}$

14. For a diamagnetic material, the dependence of the magnetic susceptibility (χ_m) on the absolute temperature (T) is given by

(1) $\chi_m \propto T$ (2) $\chi_m \propto T^{-1}$

(3) $\chi_m \propto e^T$ (4) $\chi_m \propto T^0$

15. A charge particle of charge 2 nC, mass 1 mg is projected with speed 2 m s^{-1} in xy plane as shown below. The electric field $\vec{E} = \left(2\frac{\text{N}}{\text{C}}\right)\hat{k}$ and magnetic field $\vec{B} = (2\text{ T})\hat{j}$ is applied. The acceleration of the particle is



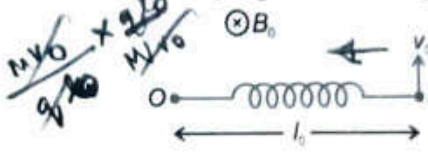
(1) $(0.018\hat{k})\text{ms}^{-2}$

(2) $(0.008\hat{k})\text{ms}^{-2}$

(3) $(0.016\hat{k})\text{ms}^{-2}$

(4) $(0.18\hat{k})\text{ms}^{-2}$

16. In a gravity free region, a point object having charge q and mass m is connected to a spring of natural length l_0 and spring constant K. A uniform magnetic field B_0 is directed into the plane of paper. The point object is given an initial velocity v_0 perpendicular to the length of spring as shown in figure



If $B_0 = \frac{mv_0}{ql_0}$, then consider the following statements and choose the correct answer from the options given below.

(I) Kinetic energy of the particle remains constant during its motion

(II) Potential energy of system remains constant

(III) Linear momentum of the particle remains constant

(IV) Angular momentum of particle about O is conserved

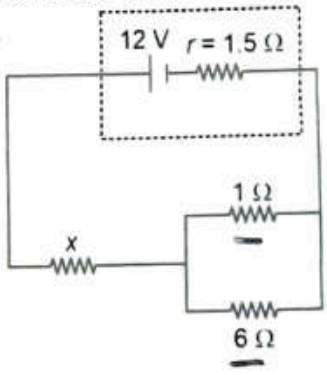
(1) Only I and II are correct

(2) Only I and IV are correct

(3) Only I, II and IV are correct

(4) Only II, III and IV are correct

17. For what value of the resistance x, the current through internal resistance is maximum?



(1) Zero

(2) $\frac{3}{14}\Omega$

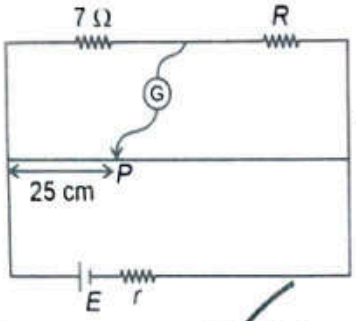
(3) $\frac{14}{3}\Omega$

(4) $\frac{3}{7}\Omega$

Space for Rough Work

18. The resistance of a conducting wire depends on
(1) Current flowing through wire
(2) Potential difference across the wire
(3) Dimensions of the conducting wire
(4) All of these

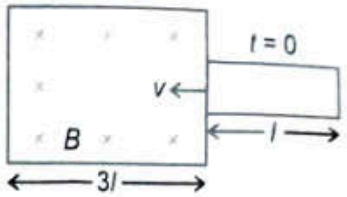
19. In an experiment to find the unknown resistance using meter bridge, the null point was found to be at 25 cm as shown. The value of resistance R is



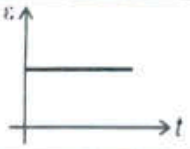
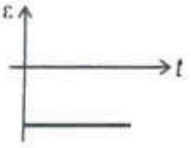
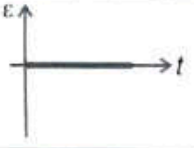
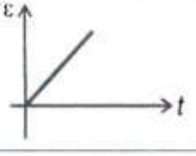
(1) 7 Ω
(2) 21 Ω
(3) 14 Ω
(4) 2.16 Ω

20. Consider the following statements and choose the correct option.
a. The intensity of magnetisation (I) is very small, negative and is directly proportional to magnetising field (H) for diamagnetic substance.
b. In non-uniform magnetic field, the paramagnetic substances are attracted towards the stronger parts of magnetic field.
c. Substances, which at room temperature, retain their ferromagnetic property for a long period of time are called permanent magnets.
(1) Only (a) and (b) are correct
(2) Only (b) and (c) are correct
(3) Only (a) and (c) are correct
(4) All (a), (b) and (c) are correct

21. A conducting loop of length l is entering inside a magnetic field region of length $3l$, and the situation shown is at $t = 0$. List-I shows time intervals and List-II shows variation of induced emf in the loop



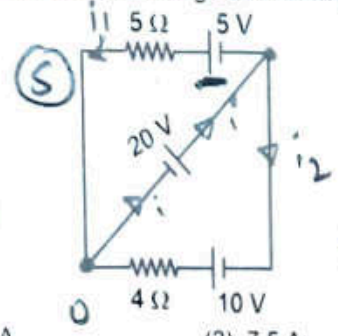
Match the List-I with List-II (Consider anticlockwise direction as positive).

	List-I		List-II
a.	$0 < t < \frac{l}{v}$	(i)	
b.	$\frac{l}{v} < t < \frac{3l}{v}$	(ii)	
c.	$\frac{3l}{v} < t < \frac{4l}{v}$	(iii)	
d.	$t > \frac{4l}{v}$	(iv)	

(1) a(i), b(iii), c(ii), d(iii)
(2) a(iii), b(i), c(ii), d(iii)
(3) a(i), b(iii), c(iii), d(ii)
(4) a(i), b(ii), c(iii), d(iii)

Space for Rough Work

22. What is the current through 20 V battery?

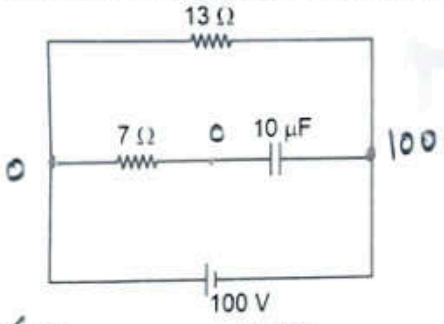


(1) 5 A (2) 7.5 A
(3) 10.2 A (4) 12.5 A

23. If current through a conductor at any instant is given by, $i = (2 + 4t)$ A where t is time in s, then the charge transferred between $t = 0$ to $t = 2$ s is

(1) 10 C (2) 12 C
(3) 5 C (4) 20 C

24. Consider a network of resistors and capacitor connected to a battery as shown. The energy stored in the capacitor in steady state condition will be



(1) 0.5 J (2) 50 J
(3) 50 mJ (4) 0.5 mJ

25. The magnetic field inside substance is B and that in free space is B_0. If $B > B_0$, then substance may be

(1) A paramagnetic material
(2) A diamagnetic material
(3) A ferromagnetic material
(4) Both (1) and (3)

26. A metal rod whose susceptibility is 549 and having cross-section area of 0.2 cm^2 is subjected to a magnetizing field of 1000 A m^{-1}. The magnetic flux produced in it is

(1) $\pi \times 10^{-10}$ Wb (2) $2.2\pi \times 10^{-7}$ Wb
(3) $4.4\pi \times 10^{-6}$ Wb (4) $2.2\pi \times 10^{-6}$ Wb

27. A thin magnetic needle oscillates in the horizontal plane in uniform magnetic field with a time period of 4 s. If the needle is broken in 2 equal parts perpendicular to its length, then time period of each part in same magnetic field will be

(1) 0.5 s (2) 1 s
(3) 1.5 s (4) 2 s

28. A paramagnetic sample shows intensity of magnetisation of 0.6 A/m, when placed in an external magnetic field of strength 1 T at a temperature 10 K. When the same sample is placed in an external magnetic field of 0.5 T at a temperature 20 K, the intensity of magnetisation is

(1) 0.6 A/m (2) 0.3 A/m
(3) 0.18 A/m (4) 0.15 A/m

29. When a 4 Ω resistance is connected to a cell, then 1 A current flows in it and when 10 Ω resistance is connected to cell, then 0.5 A current flows in it. Internal resistance of cell is

(1) 0.5 Ω (2) 2 Ω
(3) 4 Ω (4) 7 Ω

30. If the current in resistor of uniform cross-section is doubled, then consider the following statements: (Assume temperature to be approximately constant)

(i) The current density is doubled.
(ii) The electron density is doubled.
(iii) The electron density remains same.
(iv) The electron drift speed is doubled.

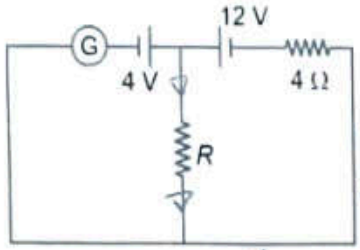
The correct statement(s) is/are

(1) Only (i) (2) Only (i) and (iii)
(3) Only (i), (ii) and (iii) (4) Only (i), (iii) and (iv)

Space for Rough Work

31. Which among the following is not a diamagnetic substance?
(1) Bismuth (Bi)
(2) Copper (Cu)
(3) Sodium chloride (NaCl)
(4) Copper chloride ($CuCl_2$)

32. In the circuit shown, the galvanometer shows zero current. The value of resistance R is (All symbols have their usual meaning)



(1) 1 Ω (2) 2 Ω
(3) 4 Ω (4) 8 Ω

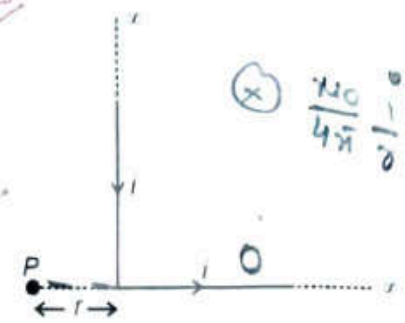
33. Given below are two statements:

Statement I: Potential difference across a battery is always less than its emf.

Statement II: Whenever current flows through a battery some energy is inevitably dissipated in its internal resistance.

In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

34. The magnetic field at point P which lies at distance r as shown in the following arrangement of current carrying wire is



(1) $\frac{\mu_0 i}{2\pi r}\otimes$ (2) $\frac{\mu_0 i}{2\pi r}\odot$
(3) $\frac{\mu_0 i}{4\pi r}\otimes$ (4) $\frac{\mu_0 i}{4\pi r}\odot$

35. Given below are two statements:

Statement I: When magnetic field (B) of a current distribution is calculated using Ampere's law, it gives the contribution to magnetic field, B only from those currents which cross the area bounded by the curve.

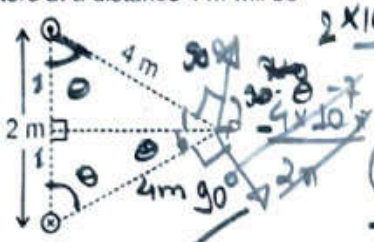
Statement II: In Ampere's law $\oint \vec{B}.d\vec{l} = \mu_0 I_{enc}$, the enclosed current includes only those currents which pierces through the area bounded by the loop.

In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

36. A wire of length 100 m is tightly wounded on a hollow tube of radius 10 mm and length 1 m. A current of 2 A is flowing in the wire. The magnetic field strength inside the tube will be
(1) 800π mT (2) 400π mT
(3) 4 mT (4) 2 mT

Space for Rough Work

37. A current of 2 A each is flowing in two long parallel wires kept at a distance of 2 m, in mutually opposite directions as shown in figure. The magnetic field induction (in μT) at a point P, equidistant from both the conductors at a distance 4 m will be

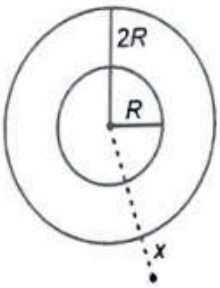


(1) 0.5 (2) 0.05
(3) 0.25 (4) 2.5

38. A long metallic hollow pipe carrying current I which is flowing parallel to its axis and has radius of cross-section r. The magnetic field

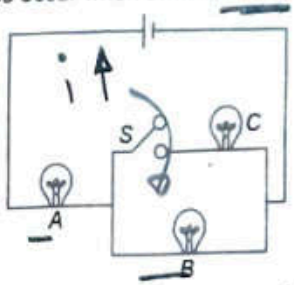
(1) At a point on its axis is zero but non-zero at the other points inside it
(2) At all points inside it is equal to zero
(3) Outside the pipe is equal to zero
(4) Directly proportional to the distance from the axis outside the pipe

39. A long hollow cylindrical conductor of inner radius R and outer radius $2R$ carries a current such that the current density is given by $J = J_0 r^2$ over its cross-section where r is distance of point from centre such that $R \le r \le 2R$. The magnetic field in the region x, where $x > 2R$ is



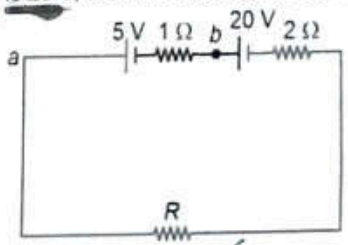
(1) $\frac{15\mu_0 J_0 R^4}{4x}$ (2) $\frac{15\mu_0 J_0 R^3}{2}$
(3) $\frac{15\mu_0 J_0 R^3}{4}$ (4) $\frac{7\mu_0 J_0 R^4}{2x^2}$

40. If A, B and C are identical lamps as shown in figure, which of the following changes to the brightness of the lamps occur when switch S is closed?



(1) Brightness of A stays same while that of B decreases
(2) Brightness of A increases while that of B decreases
(3) Brightness of A increases while that of B stays the same
(4) Brightness of A decreases while that of B increases

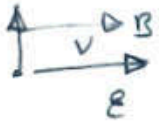
41. Two batteries one of the emf 5 V, internal resistance 1 Ω and the other of emf 20 V, internal resistance 2 Ω are connected in series with a resistance R as shown. If the potential difference between point a and b is zero, then the value of resistance of R is



(1) 1 Ω (2) 2 Ω
(3) 4 Ω (4) 0.5 Ω

Space for Rough Work

42. A charge particle is released from rest in a region of uniform electric and magnetic fields, which are parallel to each other. The locus of the particle will be



(1) A helix of constant pitch
(2) A helix of varying pitch
(3) A straight line
(4) A circle

43. A particle having charge q and mass m moves in a circular orbit of radius r with angular speed ω. The ratio of the magnitude of its angular momentum to that of its magnetic moment depends on
(1) ω, q and m
(2) ω and q
(3) ω and m
(4) q and m

44. A positive charge q is moving along positive y-axis. A uniform electric field exists towards negative x-axis. What should be the direction of magnetic field of suitable magnitude so that net force on charge is zero?
(1) Positive z-axis (2) Negative z-axis
(3) Positive x-axis (4) Negative y-axis

45. Two particles having the same specific charge $\left(\frac{q}{m}\right)$ enter a region of uniform magnetic field with the same speed but at angles of 37° and 53° with the field. Let a and b be the ratios of their pitches and radii respectively, then
(1) $a + b = \frac{25}{12}$ (2) $a - b = \frac{7}{12}$
(3) $ab = 1$ (4) All of these

[CHEMISTRY]

46. Consider the given statements
a. $[Fe(CN)_6]^{3-}$ has magnetic moment of a single unpaired electron.
b. $[Co(H_2O)_6]^{3+}$ is paramagnetic in nature and is an outer orbital complex.
c. $[Co(C_2O_4)_3]^{3-}$ is an inner orbital complex.
d. $[Ti(H_2O)_6]^{3+}$ is a colourless species.
The correct statements are
(1) (a) and (b) only (2) (a), (b) and (c) only
(3) (a) and (c) only (4) (a), (c) and (d) only

47. Given below are the two statements
Statement I: Total number of stereoisomers of $[PtCl_2(en)_2]^{2+}$ is 4
Statement II: cis-$[PtCl_2(en)_2]^{2+}$ is optically active
In the light of above statements choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

48. Match List-I with List-II

	List-I Ion		List-II Spin only magnetic moment in BM
a.	Fe^{3+}	(i)	3.87
b.	Co^{2+}	(ii)	4.90
c.	Cr^{2+}	(iii)	1.73
d.	Ti^{3+}	(iv)	5.9

The correct match is
(1) a(iv), b(i), c(ii), d(iii) (2) a(iv), b(i), c(iii), d(ii)
(3) a(ii), b(i), c(iv), d(iii) (4) a(i), b(iv), c(ii), d(iii)

Space for Rough Work

49 Correct order of negative hydration of the given ions is

(1) $V^{2+} > Cr^{2+} > Co^{2+} > Ni^{2+}$

(2) $Ni^{2+} > Co^{2+} > Cr^{2+} > V^{2+}$

(3) $Co^{2+} > Ni^{2+} > Cr^{2+} > V^{2+}$

(4) $Ni^{2+} > Cr^{2+} > Co^{2+} > V^{2+}$

50 Consider the following statements

(a) The highest Mn fluoride is MnF_6 whereas highest Mn oxide is Mn_2O_7.

(b) V_2O_3 is basic while V_2O_5 is amphoteric in nature.

(c) V_2O_5 reacts with alkalies and gives VO_4^{3-} ions.

(d) CrO and Cr_2O_3 are basic oxides of chromium.

The correct statements are

(1) (a), (b) and (c) only (2) (b) and (c) only

(3) (b), (c) and (d) only (4) (a), (b), (c) and (d)

51 Given below are the two statements

Statement I: $KMnO_4$ on heating at 513 K gives K_2MnO_4, Mn_2O_3 and O_2 as products.

Statement II: MnO_4^{2-} in acidic solution disproportionates to MnO_4^- and MnO_2.

In the light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

52 Correct order of ligand field strength of the given species based on spectrochemical series is

(1) $C_2O_4^{2-} > F^- > S^{2-} > I^-$

(2) $S^{2-} > C_2O_4^{2-} > F^- > I^-$

(3) $S^{2-} > C_2O_4^{2-} > I^- > F^-$

(4) $C_2O_4^{2-} > S^{2-} > F^- > I^-$

53 For a first order reaction A → Products, initial concentration of A is 0.20 M which becomes 0.0125 M after 20 minutes. Half-life of the reaction is nearly

(1) 480 s (2) 260 s

(3) 61 s (4) 301 s

54 Given below are the two statements

Statement I: Order is applicable only to elementary reactions.

Statement II: Molecularity of a reaction cannot be zero or a non-integer.

In the light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

55 $3A \xrightarrow{k} B$ is a zero order reaction, where rate constant (k) = 2×10^{-2} mol L^{-1} s^{-1}. If the initial concentration of A is 2.4 then the time taken to complete 80% of the reaction will be

(1) 96 s (2) 48 s

(3) 32 s (4) 16 s

56 If the rate of reaction quadruples when temperature changes from 27°C to 47°C then the energy of activation for the reaction will be

(1) 27.2 kJ mol^{-1} (2) 61.3 kJ mol^{-1}

(3) 55.1 kJ mol^{-1} (4) 12.8 kJ mol^{-1}

57 For the reaction, $A + 3B \rightarrow 2C$, if $\frac{d[C]}{dt} = 4 \times 10^{-3}$ mol L^{-1} s^{-1} then the value of $-\frac{d[B]}{dt}$ will be

(1) 4×10^{-3} mol L^{-1} s^{-1} (2) 8×10^{-3} mol L^{-1} s^{-1}

(3) 2×10^{-4} mol L^{-1} s^{-1} (4) 6×10^{-3} mol L^{-1} s^{-1}

58 The lanthanoid ion having 2 unpaired electrons is

(1) Er^{2+} (2) Tm^{2+}

(3) Nd^{2+} (4) Dy^{2+}

Space for Rough Work

59. Consider the following complexes

(a) $[CoF_6]^{3-}$ (b) $[Co(NH_3)_6]^{3+}$

(c) $[Co(en)_3]^{3+}$ (d) $[Co(H_2O)_6]^{3+}$

The correct order of [redacted] of light absorbed for the given complexes is

(1) (c) > (b) > (d) > (a) (2) (a) > (d) > (c) > (b)

(3) (a) > (b) > (d) > (c) (4) (a) > (d) > (b) > (c)

60. Match List-I with List-II

	List-I Species		List-II Colour
a.	$K_2Cr_2O_7$	(i)	Green
b.	Mn_2O_7	(ii)	Orange
c.	MnO_4^-	(iii)	Yellow
d.	CrO_4^{2-}	(iv)	Purple

The **correct** match is

(1) a(iii), b(i), c(iv), d(ii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(ii), c(iv), d(i)

61. Given below are the two statements

Statement I: Potassium dichromate is used as a [redacted] standard in volumetric analysis.

Statement II: One mole of dichromate ions oxidises [redacted] moles of Fe^{2+} ions in acidic medium.

In the light of above statements, choose the correct answer.

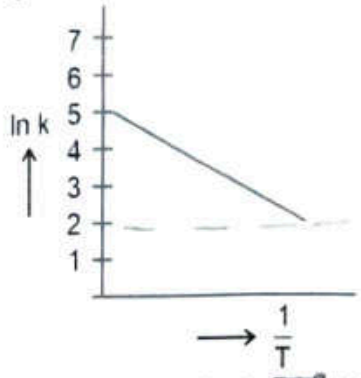
(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

62. For an elementary chemical reaction, the Arrhenius plot is given below



If the energy of activation is [redacted] kJ mol^{-1} and $R = 8.3\ JK^{-1}\ mol^{-1}$ then the temperature at which the rate constant becomes $e^3\ min^{-1}$, is

(1) 350 K (2) 450 K

(3) 280 K (4) 320 K

63. Consider the given statements about catalyst

(a) It does not alter Gibbs energy, ΔG of a reaction.

(b) It does not catalyse non-spontaneous reaction.

(c) It helps in attaining the equilibrium faster by increasing the equilibrium constant of the reaction.

(d) It catalyses the forward as well as the backward reactions to the same extent.

The correct statement(s) is/are

(1) (b) and (c) only (2) (a) and (d) only

(3) (a) only (4) (a), (b) and (d) only

64. The reaction, $A_2 + B_2 \rightarrow 2AB$ follows the given mechanism.

$B_2 \rightleftharpoons B + B$ (fast)

$A_2 + B \rightarrow AB + A$ (slow)

$A + B \rightarrow AB$ (fast)

The overall order of the reaction will be

(1) 1 (2) 0.5

(3) 2 (4) 1.5

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65. The rate of the first order reaction is 1.0×10^{-2} Ms^{-1} at 60 seconds and 0.5×10^{-2} Ms^{-1} at 120 seconds after the initiation of the reaction. The rate constant of the reaction is

(1) $2.41 \times 10^{-3}\ s^{-1}$ (2) $1.15 \times 10^{-2}\ s^{-1}$

(3) $4.25 \times 10^{-3}\ s^{-1}$ (4) $6.18 \times 10^{-2}\ s^{-1}$

66. Match List-I with List-II

	List-I		List-II
a.	Zero order	(i)	Decomposition of H_2O_2 catalysed by iodide ion in an alkaline medium
b.	First order	(ii)	Artificial radioactive decay
c.	Pseudo first order	(iii)	Decomposition of gaseous ammonia on a hot platinum surface at high pressure
d.	Second order	(iv)	Inversion of cane sugar in acidic medium

The **correct** match is

(1) a(iii), b(i), c(iv), d(ii) (2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(iv), c(i), d(iii) (4) a(iii), b(ii), c(iv), d(i)

67. Correct order of basic strength of the given compound is

(1) $Eu(OH)_3 > Pm(OH)_3 > Tb(OH)_3 > Er(OH)_3$

(2) $Pm(OH)_3 > Eu(OH)_3 > Tb(OH)_3 > Er(OH)_3$

(3) $Pm(OH)_3 > Tb(OH)_3 > Eu(OH)_3 > Er(OH)_3$

(4) $Tb(OH)_3 > Er(OH)_3 > Eu(OH)_3 > Pm(OH)_3$

68. Given below are the two statements, one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A): $[MnBr_4]^{2-}$ has spin only magnetic moment value of 5.9 BM.

Reason (R): $[MnBr_4]^{2-}$ has 5 unpaired electrons.

In light of above statements, choose the **correct** answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

69. The pair of ions which give colour in aqueous medium is

(1) Mn^{3+} and Zn^{2+} (2) Sc^{3+} and V^{3+}

(3) Fe^{2+} and Co^{3+} (4) Ti^{4+} and Ni^{2+}

70. Given below are the two statements

Statement I: Half-life of a zero order reaction is directly proportional to the initial concentration of the reactants.

Statement II: Unit of rate constant for a second order reaction is $mol^2 L^{-2} s^{-1}$

In light of above statements, choose the correct answer.

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

71. Crystal field stabilization energy for the complex $[CoF_6]^{3-}$ will be (Ignore pairing energy)

(1) $-1.6\Delta_0$ (2) $-0.6\Delta_0$

(3) $-2.4\Delta_0$ (4) $-0.4\Delta_0$

72. Match List-I with List-II

	List-I Coordination compounds		List-II Metal present
a.	Haemoglobin	(i)	Rhodium
b.	Cyanocobalamine	(ii)	Magnesium
c.	Chlorophyll	(iii)	Cobalt
d.	Wilkinson catalyst	(iv)	Iron

The **correct** match is

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(1) a(iii), b(iv), c(ii), d(i) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iv), b(i), c(ii), d(iii) (4) a(iv), b(iii), c(ii), d(i)

73. For the reaction, $A(g) \rightarrow 2B(g) + C(g)$, if the initial pressure of A is 320 mm of Hg and the rate constant of the reaction is $2 \times 10^{-3}\ s^{-1}$ then the time at which the total pressure of the reaction mixture becomes 920 mm of Hg will be approximately (log 2 = 0.3010)
(1) 1874 s (2) 945 s
(3) 1386 s (4) 1264 s

74. Given below are the two statements; one is labelled as Assertion (A) and other is labelled as Reason (R)
Assertion (A): The geometrical isomers of $[Fe(NH_3)_2(CN)_4]^-$ are optically inactive
Reason (R): Both geometrical isomers of $[Fe(NH_3)_2(CN)_4]^-$ possess plane of symmetry.
In light of above statements, choose the correct answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

75. Correct order of first ionisation enthalpy of the given species is
(1) Fe > Mn > Cr > V (2) Mn > Fe > Cr > V
(3) Mn > V > Fe > Cr (4) Fe > Mn > V > Cr

76. Most common oxidation state of vanadium and nickel respectively are
(1) +5 and +2 (2) +4 and +2
(3) +5 and +4 (4) +3 and +4

77. An excess of $AgNO_3$ is added to 200 mL of 0.4 M solution of tetraammineaquachloridocobalt (III) chloride.
The number of moles of AgCl precipitated would be
(1) 0.08 (2) 0.16
(3) 0.24 (4) Zero

78. Which of the given carbonyls will have the weakest C–O bond?
(1) $[Co(CO)_4]^-$ (2) $[Ni(CO)_4]$
(3) $[Cr(CO)_6]$ (4) $[Mn(CO)_6]^+$

79. Incorrect statement among the following is
(1) Tb^{4+} has half-filled *f*-orbitals and acts as an oxidant.
(2) Ce(IV) is a good analytical reagent.
(3) In mischmetal, the percentage of lanthanoid metal is about 95%.
(4) Lanthanoid metals do not react with dilute acids.

80. Which among the following has the highest value of enthalpy of atomisation?
(1) V (2) Cr
(3) Mn (4) Co

81. Consider the following statements about interstitial compounds
(a) They are chemically inert
(b) They act as insulator
(c) They have high melting points
(d) They are usually nonstoichiometric
The correct statements are
(1) (a), (c) and (d) only (2) (a) and (c) only
(3) (a), (b) and (d) only (4) (a), (b), (c) and (d)

82. Match List-I with List-II

	List-I Species		List-II Use
a.	V_2O_5	(i)	Used in the treatment of lead poisoning
b.	EDTA	(ii)	Catalyses the oxidation of SO_2 in manufacture of H_2SO_4
c.	cis-platin	(iii)	Production of NH_3 from N_2 and H_2 mixture
d.	Iron catalysts	(iv)	Inhibits the growth of tumours

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Choose the **correct** match

(1) a(ii), b(i), c(iv), d(iii) (2) a(i), b(ii), c(iv), d(iii)

(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(iv), c(i), d(ii)

83. Which among the following is an example of homoleptic complex?

(1) Dichloridobis(ethane-1,2-diamine)cobalt (III) chloride

(2) Pentaamminecarbonatocobalt (III) chloride

(3) Potassium trioxalatochromate (III)

(4) Diamminechloridonitrito-N-platinum (II)

84. Which among the following is the correct electronic configuration of the central atom in $[MnCl_6]^{3-}$?

(1) $t_{2g}^3 e_g^1$ (2) $t_{2g}^4 e_g^0$

(3) $e^2 t_2^2$ (4) $e^4 t_2^0$

85. **Incorrect** match among the following is

(1)	$[NiCl_4]^{2-}$	–	Tetrahedral and paramagnetic
(2)	$[Ni(CO)_4]$	–	Square planar and diamagnetic
(3)	$[PtCl_4]^{2-}$	–	Square planar and diamagnetic
(4)	$[Ni(CN)_4]^{2-}$	–	Square planar and diamagnetic

86. Given below are the two statements

Statement I: NO_2^- and SCN^- are ambidentate ligands.

Statement II: Linkage isomerism arises in a coordination compound containing ambidentate ligand.

In light of above statements, choose the correct answer.

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are correct

(3) Both Statement I and Statement II are incorrect

(4) Statement I is incorrect but Statement II is correct

87. Jahn-Teller effect is not observed in the high spin octahedral complexes of

(1) d^4 (2) d^6

(3) d^7 (4) d^5

88. In a zero-order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 20°C to 70°C then the rate of the reaction will become

(1) 60 times (2) 128 times

(3) 32 times (4) 16 times

89. During the kinetic study of the reaction

$A + B \rightarrow C$, following results were obtained

Experiment	[A] M	[B] M	Rate (Ms^{-1})
(i)	0.1	0.2	1×10^{-5}
(ii)	0.2	0.4	8×10^{-5}
(iii)	0.2	0.8	32×10^{-5}

Based on the above data, the order of reaction with respect to A and B respectively are

(1) 1 and 1 (2) 0.5 and 2

(3) 2 and 0 (4) 1 and 2

90. Give below are the two statements

Statement I: Thorium is a transuranium element.

Statement II: U^{3+} is larger in size as compared to Am^{3+}.

In light of above statements, choose the correct answer.

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are correct

(3) Both Statement I and Statement II are incorrect

(4) Statement I is incorrect but Statement II is correct

Space for Rough Work

[BOTANY]

91. Which of the following traits can be expressed by female *Drosophila* in both homozygous as well as heterozygous conditions?
(1) Miniature wing
(2) White eye colour
(3) Brown body colour
(4) All the sex-linked traits

92. Which of the following is not observed in phenylketonuria?
(1) Mental retardation
(2) Reduced hair pigmentation
(3) Reduction in skin pigmentation
(4) Accumulation of pyruvic acid in brain

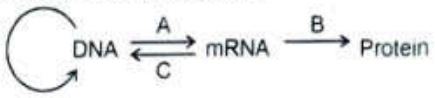
93. Thalassemia differs from sickle cell anaemia because the former
(1) Cannot be studied via pedigree analysis
(2) Is an autosomal recessive disorder
(3) Can affect an individual by causing anaemia
(4) Is caused due to synthesising too few globin molecules

94. Affected couple may still produce unaffected children in case of
A. Myotonic dystrophy B. Colour blindness
C. Down's syndrome D. Turner's syndrome
The **correct** ones is/are
(1) A and C only (2) C only
(3) All except B (4) D only

95. Who suggested that in order to code for all 20 amino acids, the code should be made up of 3 nucleotides?
(1) H.G. Khorana (2) M. Nirenberg
(3) G. Gamow (4) Severo Ochoa

96. Presence of a Y chromosome plays a role in determination of fertile males in which of the following pairs of species?
(1) Human and *Drosophila*
(2) *Drosophila* and Grasshopper
(3) Human and Honeybee
(4) Grasshopper and Honeybee

97. Observe the following schematic representation of flow of genetic information.



Read the following statements regarding the above schematic representation and choose the **incorrect** one.
(1) The processes A and B use the concept of complementarity to explain the relationship between their template and product.
(2) The process C can use the entire length of the template molecule.
(3) The processes A and C would require polymerases.
(4) The process C has been observed in certain acellular infectious agents.

98. Which of the following genotypes in human beings results skin colour slightly darker than intermediate, assuming external environment is not altering the phenotype?
(1) AaBbcc (2) AAbbcc
(3) AaBbCC (4) AABBCC

99. Gametes with which of the following genotypes can be formed by a plant having genotype TtRryy by independent assortment?
(1) try (2) TrR
(3) TrY (4) tRY

Space for Rough Work

100. If an *E. coli* cell containing heavy DNA ($^{15}N^{15}N$) is allowed to multiply in ^{14}N medium for 40 minutes, what percentage of daughter cells would possess light DNA ($^{14}N^{14}N$)?

(1) Zero (2) 50%

(3) 25% (4) 100%

101. The technique of DNA fingerprinting was initially developed by

(1) Alec Jeffreys (2) H.G. Khorana

(3) Marshall Nirenberg (4) Jacob and Monod

102. Consider the following statements.

Allelic sequence variation has traditionally been described as a A if more than one variant/allele at a locus occurs in human population with a frequency greater than B .

Choose the option that **correctly** fills the blanks A and B.

	A	B
(1)	Satellite DNA	0.1
(2)	DNA polymorphism	0.01
(3)	Snips	1.0
(4)	Satellite DNA	0.001

103. The sequence of nucleotides in the coding region of mRNA (excluding sequence of stop codon) can be reliably determined if the polypeptide is made of

(A) Two monomeric units, Met and Trp

(B) One monomeric unit, any amino acid

(C) Two monomeric units, Val and Glu

Choose the **correct** answer from the following options.

(1) A only (2) A and B

(3) B only (4) B and C only

104. The plant *Vicia faba*, lacks all of the following in its genetic material, **except**

(1) Tritiated thymidine

(2) Deoxyuridine

(3) Deoxyribose

(4) Cytidine

105. How many complete turns would be present in a stretch of human DNA molecule formed by 150 nucleotide pairs?

(1) No more than 15 (2) 25

(3) More than 25 (4) 16

106. Read the following statements and state **true (T)** or **false (F)** to them and choose the **correct** option.

(A) All the three types of RNA polymerases in nucleus of eukaryotes are involved in transcription of genes.

(B) Most eukaryotic cistronic genes contain information for a single polypeptide but not in a single continuous base sequence.

(C) Most prokaryotic genes contain information of polypeptide in a single continuous base sequence but can possesses information of more than one polypeptide.

(D) Frame shift mutation in introns, that does not affect splicing pattern of hnRNA, also does not affect protein synthesis.

	A	B	C	D
(1)	T	T	T	T
(2)	T	F	T	T
(3)	F	T	F	T
(4)	T	T	F	T

107. Which of the following codons in eukaryotes has dual functions regarding translation?

(1) AUG (2) UGG

(3) UGA (4) GAG

Space for Rough Work

108. Select the **incorrect** statement

(1) In post-transcriptional processing within eukaryotic cells, methyl guanosine triphosphate is added at 5' end of hnRNA.

(2) Frederick Griffith, in a series of experiments with *Streptococcus pneumoniae*, proved that DNA was the 'transforming principle'.

(3) In DNA replication, the discontinuously synthesized fragments are later joined by the enzyme DNA ligase.

(4) RNA was the first genetic material.

109. Mendel chose seven pairs of traits of garden pea for experiment. If a pea plant hybrid for both flower position and pod colour self-pollinates, what would be the probability of formation of green pods at the axial position in the same plant post fertilization?

(1) 1 (2) $\frac{9}{16}$

(3) $\frac{3}{16}$ (4) $\frac{1}{4}$

110. In sickle cell anemia, a mutation leads to replacement of the amino acid at the 6th position of the beta globin chain of hemoglobin by another amino acid.

Which of the following is **correct** regarding the above statement?

(1) Valine replaces Glutamic acid

(2) Glutamic acid replaces Valine

(3) Phenylalanine replaces Tyrosine

(4) Tyrosine replaces Phenylalanine

111. The male grasshopper (2n = 23) differs from the male *Drosophila* in

(1) The number of X chromosomes found in each diploid cell

(2) Forming 50% gametes having X-chromosome

(3) The total lack of Y chromosome

(4) The role played by heterogametic parent in sex determination

112. Select the **odd** one out w.r.t. male heterogamety from the following options.

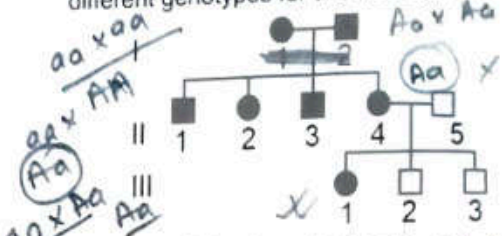
(1) Chickens

(2) Grasshoppers

(3) *Drosophila*

(4) Humans

113. The following pedigree depicts the inheritance of a genetic disorder in three generations of a family. Individuals I-2 and II-3 are detected to possess different genotypes for the disorder.



Read the following statements and choose the **correct** one.

(1) The disorder cannot be colour blindness

(2) The disorder can be myotonifc dystrophy, the probability of III-1 being homozygous is $\frac{1}{2}$

(3) The disorder can be phenylketonuria and the probability of III-3 having the recessive allele is $\frac{1}{2}$

(4) The disorder is autosomal in nature and can be dominant or recessive

114. The genome of $\phi \times 174$ phage is a polymeric structure composed of 5386 units.

In the given structure this number is not said to be true for

(1) Glycosidic linkages

(2) Dinucleotides

(3) Nitrogeneous bases

(4) Phosphodiester linkages

Space for Rough Work

115. Read the following statements:

Statements-I: During translation, the interaction between the adapter molecule and template is determined by complementarity.

Statement-II: The 5' end of tRNA is amino acid acceptor end.

Choose the **correct** option w.r.t. the above statements.

(1) Both statements I and II are incorrect
(2) Both statements I and II are correct
(3) Only statement I is correct
(4) Only statement II is correct

116. Select the **correctly** matched pair from the following options.

(1) Severo Ochoa enzyme – Not a polymerising enzyme
(2) 23S rRNA – Peptidyl transferase
(3) DNA helicase – Uncoils DNA, only during transcription
(4) Restriction endonuclease – Is not used in DNA fingerprinting

117. Match the following list-I with list-II w.r.t. human genome.

	List-I		List-II
(a)	Genes that have yet unknown functions	(i)	Less than 2 per cent
(b)	Genomic content that codes for protein	(ii)	More than 99 per cent
(c)	Nucleotide sequence identical across the human population	(iii)	Less than 0.1 per cent
(d)	Base sequence that may vary in the genomes of two people	(iv)	More than 50 per cent

Choose the most appropriate option.

(1) a-iv, b-i, c-ii, d-iii (2) a-i, b-iv, c-iii, d-ii
(3) a-ii, b-iv, c-i, d-iii (4) a-ii, b-i, c-iii, d-iv

118. Read the following assertion and reason carefully.

Assertion (A): There is independent assortment of two genes either they are tightly linked or present on the homologous chromosomes.

Reason (R): Two genes situated at the same loci of homologous chromosomes are considered to be a pair of alleles.

Choose the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is false but (R) is true
(4) Both (A) and (R) are false

119. In the F_2 generation of a typical Mendelian dihybrid cross, the proportion of progeny homozygous for one character and heterozygous for the other character is

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$
(3) $\frac{3}{8}$ (4) $\frac{9}{16}$

120. Match the following **Column-I** with **Column-II**

	Column-I		Column-II
(a)	Discovery of X-body in insect	(i)	Alfred Sturtevant
(b)	Graphical representation to show all possible genotypes of offspring in a cross	(ii)	T.H. Morgan
(c)	Experimental verification of chromosomal theory of inheritance	(iii)	Reginald C. Punnett
(d)	Mapped the position of genes on chromosome	(iv)	Henking

Choose the **correct** option.

Space for Rough Work

(1) a-iv, b-iii, c-i, d-ii (2) a-iv, b-ii, c-iii, d-i
(3) a-iv, b-iii, c-ii, d-i (4) a-iii, b-iv, c-ii, d-i

121. Assuming that each gene governs the expression of a different character and follow the Mendelian inheritance, the phenotypic ratio produced in the progeny of the following cross would be

TtRr × ttrr

(1) 1 : 2 : 1 (2) 9 : 3 : 3 : 1
(3) 1 : 1 : 1 : 1 : 1 : 1 (4) 1 : 1 : 1 : 1

122. Expressed sequence tags are referred as
(1) Identifying all the genes that are expressed as RNA
(2) Sequencing only the non-coding sequences of the genome
(3) Sequencing the whole set of genome
(4) Identifying the locations of repeated sequences in the genome

123. The number of structural genes, mRNAs and different enzymes formed by a *lac* operon in the presence of inducer are respectively
(1) 3, 1 and 3 (2) 3, 2 and 3
(3) 3, 4 and 3 (4) 1, 1 and 4

124. The entities represented by terms 'Nuclein' and 'Nucleoid' can be present together in
(1) *E. coli* (2) TMV
(3) *Vicia faba* (4) QB phage

125. The number of nucleosomes in the 1.36 mm long *E. coli* genome would be
(1) Zero
(2) 2.3×10^4
(3) One
(4) Two, including plasmid

126. Which of the following is **not** formed by RNA polymerase I in eukaryotes?
(1) 16S rRNA (2) 5.8S rRNA
(3) 28S rRNA (4) 18S rRNA

127. Choose the option that **correctly** classifies the following organisms based on the number of X chromosomes in them.

A. An individual with Turner's syndrome
B. Female grasshopper
C. A female with Down's syndrome
D. An individual with Klinefelter's syndrome

	1 X chromosome	2 X chromosomes
(1)	A, B, D	C
(2)	A, D	B, C
(3)	A	B, C, D
(4)	B	A, C, D

128. If a species contains only 20 types of amino acids, the minimum number of sense codons required to code for them would be
(1) 64 (2) 61
(3) 21 (4) 20

129. From the following select the codon that is **odd** one out.
(1) UAA (2) UAU
(3) UAG (4) UGA

130. Who amongst the following scientists, credited for developing method for determination of amino acid sequences in proteins?
(1) Taylor (2) Alec Jeffreys
(3) Frederick Sanger (4) Frederick Griffith

131. Chromosomal theory of inheritance was proposed by
(1) Sutton (2) Meischer
(3) Flemming (4) Mendel

132. Law of dominance can be used to explain
(A) Formation of exclusively tall pea plants in the progeny of pure tall and dwarf parents.
(B) Formation of more gametes having T allele in the F_1 progeny as compared to gametes having t allele.
(C) Formation of pink flowered progeny upon selfing a red flowered *Antirrhinum* plant.

The **correct** one(s) is/are
(1) C only (2) A and B only
(3) A only (4) B and C only

Space for Rough Work

133. Read the following statements carefully.

A. Klinefelter's syndrome and haemophilia are frequent in females but due to different reasons.

B. In honey bee, drones do not receive any genes from their male parent.

C. The intensity of thalassemia is directly proportional to the quantity of alleles that are unaffected.

Choose the **correct** option

	A	B	C
(1)	F	T	T
(2)	T	F	T
(3)	T	F	F
(4)	F	T	F

134. Sahiwal cows are

(1) Not developed from wild cows

(2) Domesticated for desired traits

(3) Devoid of variations

(4) Formed by exploitation of asexual reproduction

135. Mendel published his research in the year

(1) 1822

(2) 1865

(3) 1901

(4) 1902

[ZOOLOGY]

136. The Reproductive and Child Health care (RCH) programmes aim to improve the overall reproductively healthy society. Which of the following is **not** an objective of these programmes?

(1) Creating awareness among youth about the myths and misconceptions regarding sex related aspects

(2) Implementing a total ban on the use of all the contraceptive methods to increase the birth rates

(3) Providing professional expertise and material support for better post-natal care of the mother

(4) Spreading awareness about "one child norm" *via* various media platforms

137. Complete the analogy by selecting the correct option w.r.t. STIs and their causative organisms.

Genital herpes : Viruses : : ____ : Bacteria

(1) Trichomoniasis (2) Hepatitis-B

(3) AIDS (4) Chlamydiasis

138. Choose the **odd** one w.r.t. the contraceptives that employ the use of hormones in their mode of action.

(1) Progestasert

(2) LNG-20

(3) Multiload-375

(4) Implants

139. Select the type of natural selection/event that was observed in industrial melanism in England and its effect on the moth population in 1920s.

(1) Stablising selection, maintains the average phenotype by favouring the intermediate grey color phenotype.

(2) Directional selection, shifts the mean of the character in one specific direction *i.e.*, towards the dark coloured phenotype.

(3) Disruptive selection, favours both extreme phenotypes while eliminating the intermediate ones.

(4) Genetic drift, change in allele frequency purely due to a random chance event resulting in increase in number of white phenotype.

Space for Rough Work

140. Which of the following statements correctly represents a core principle of Lamarck's evolutionary theory?

(1) Evolution is driven by random genetic mutations that occur during gamete formation

(2) Continuous disuse of an organ over many generations, leads to its gradual degeneration, resulting in a vestigial organ

(3) Only the individuals with the most favourable variations survive and reproduce in a population

(4) New species arise *via* sudden, large-scale variations which are called saltations

141. How many of the organisms given in the box below are placental mammals?

Lemur, Spotted cuscus, Bobcat, Flying squirrel, Numbat, Anteater

Select the **correct** option.

(1) Four (2) Two

(3) Three (4) Five

142. Choose the **incorrect** match.

(1)	Amniocentesis	–	Used to detect chromosomal abnormalities in foetus
(2)	Lactational amenorrhea	–	Natural method of contraception based on the absence of ovulation during intense lactation
(3)	Diaphragm	–	Reusable barrier used for conception that also protects user from STI
(4)	Cu7	–	Suppresses sperm motility and fertilising capacity of sperms

143. **Assertion (A):** The progestogens alone pills are effective as emergency contraceptives.

Reason (R): These pills are used to avoid possible pregnancy when taken within 72 hours of unprotected coitus.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) correctly explains (A)

(2) Both (A) and (R) are true, (R) does not explain (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

144. In a standard 28-days menstrual cycle, the "fertile window" during which a couple should avoid coitus to prevent conception is

(1) Day 1 to 5 (2) Day 6 to 9

(3) Day 10 to 17 (4) Day 18 to 28

145. Consider the given statements w.r.t. STIs and select the **incorrect** one.

(1) The incidences of these infections is particularly high among individuals in the age group of 15-24 years.

(2) Potential late complications of untreated STIs are PID and ectopic pregnancies.

(3) Infected females may often be asymptomatic and hence, may remain undetected for long.

(4) HIV infections and genital herpes can be cured completely, if detected early and treated properly.

146. In India, a 22 weeks pregnant woman seeks an abortion after doctor identifies a substantial fetal abnormality. Proceeding with the termination of pregnancy at this specific stage

(1) Opinion of only one registered medical practitioner is required

(2) Opinion of atleast two registered medical practitioners is required

(3) Mandatory approval from a state level medical board is needed

(4) Termination is strictly prohibited as the pregnancy has crossed 20 weeks

Space for Rough Work

147. Which of the following is the set of contraceptive methods that work solely on the principle of reducing the chances of fertilisation by preventing the meeting of gametes without the use of contraceptives or devices?
(1) Lippes loop, Multiload-375, LNG-20
(2) Periodic abstinence, Coitus interruptus, Lactational amenorrhea
(3) Vaults, Diaphragms, Cervical caps
(4) Saheli, Mala-D, Implants

148. In year 1953, S.L. Miller successfully synthesized organic molecules by simulating primitive Earth conditions. Which of the following correctly identifies the parameter he used?
(1) CH_4, NH_3, O_2 and water vapour at 800°C with UV radiations
(2) CH_4, NH_3, H_2, and water vapour at 80°C with electric discharge
(3) CH_4, NH_3, H_2, and water vapour at 800°C with electric discharge
(4) CO_2, NH_3, H_2, and water vapour at 100°C with electric discharge

149. In a stable population of 2000 individuals, 320 individuals are found to be homozygous recessive for a specific trait. What would be the frequency of the heterozygous individuals in this population? (Assume the population to be in Hardy-Weinberg equilibrium)
(1) 0.48 (2) 0.32
(3) 0.16 (4) 0.80

150. Consider the following statements regarding the history of life forms and identify the one that is **incorrect**.
(1) The first non-cellular life forms originated about 3 billion years ago
(2) Lobefins evolved into the first amphibians that could live on both land and water
(3) *Tyrannosaurus rex*, a herbivorous dinosaur that lived in herds was about 12 feet tall
(4) About 65 million years ago, dinosaurs suddenly disappeared from the Earth

151. Match the plant forms in Column I with their direct descendants in Column II and select the **correct** option.

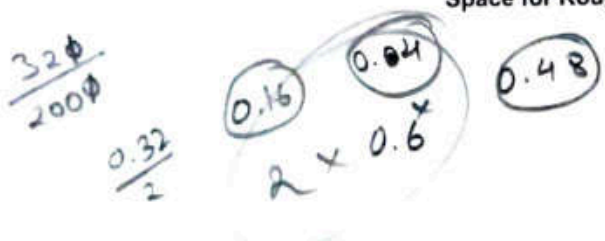
	Column I		Column II
a.	Chlorophyte ancestors	(i)	Seed ferns
b.	Tracheophyte ancestors	(ii)	Gnetales
c.	Psilophyton	(iii)	Zosterophyllum
d.	Progymnosperms	(iv)	Bryophytes

(1) a(iii), b(i), c(iv), d(ii) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(ii), c(i), d(iv)

152. Which of the following statements is **incorrect**?
(1) The original variety of Darwin's finches were seed-eaters that evolved into insectivorous and vegetarian forms
(2) Australian marsupials evolved from a single ancestral stock within the isolated Australian continent
(3) When more than one adaptive radiation occurs in an isolated geographical area (representing different habitat), it represents convergent evolution
(4) Placental mammals in Australia do not show any similarity to the corresponding Australian marsupials

153. Read the following statements and select the **correct** option.
Statement (A): The thorns of *Bougainvillea* and tendrils of *Cucurbita* represent homology, indicating that these two plants share a common ancestry.
Statement (B): The wings of a butterfly and the wings of a bird are analogous structures that have evolved due to divergent evolution to perform similar functions.
(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Statement (A) is correct but statement (B) is incorrect
(4) Statement (A) is incorrect but statement (B) is correct

Space for Rough Work



154. The flippers of penguins and that of dolphins
(1) Are homologous organs that have evolved due to divergent evolution
(2) Represent homology and share a similar anatomical bone structure
(3) Are analogous structures that have evolved due to convergent evolution
(4) Are vestigial organs that no longer perform any significant function

155. Based on the embryological studies of different vertebrates, select the statement that accurately reflects the findings of Karl Ernst von Baer that led to the disproval of the "Biogenetic Law".
(1) Embryos of higher vertebrates pass through the adult stages of lower vertebrates like fish
(2) All vertebrate embryos develop a functional row of gill slits that are retained in adults
(3) Embryos never pass through the adult stages of other animals; instead, they share certain common embryonic features that diverge during development
(4) Ontogeny always recapitulates phylogeny, providing a linear record of a species evolutionary history

156. A childless couple sought medical advice as the female partner was unable to produce functional ova, but had a healthy uterus and fallopian tubes, capable of supporting a pregnancy. The doctor recommended a technique where an ovum from a donor can be transferred into her fallopian tube. Identify this *in-vivo* fertilisation method and choose the **correct** option.
(1) ZIFT (2) GIFT
(3) IUT (4) ICSI

157. Arrange the following in the correct ascending order w.r.t. their brain capacities and select the correct option.
(A) *Homo erectus*
(B) *Homo habilis*
(C) *Homo neanderthalensis*
(D) *Australopithecus*
(1) D → B → A → C (2) B → D → A → C
(3) A → C → D → B (4) D → A → C → B

158. Select the **incorrect** match regarding evolution of different life-forms through geological time scale.

	Column I (Time Period)		Column II (Major Event)
(1)	350 mya	–	Evolution of jawless fish
(2)	320 mya	–	Sea weeds and a few plants existed
(3)	2000 bya	–	Origin of first cellular life forms
(4)	1.5 mya	–	Existence of *Homo erectus*

159. Choose the option that **incorrectly** represents the feature or role of the given contraceptives.

(1)	Saheli	–	Non-steroidal oral pill taken once a week
(2)	Lippes loop	–	Non-medicated IUD that prevents implantation
(3)	Cu7	–	Hormone releasing IUD that makes cervix hostile to sperms
(4)	Vasectomy	–	Terminal method with very poor reversibility

160. Select the **incorrect** statement.
(1) While all RTIs are generally localized to the reproductive tract, certain STIs like AIDS and Hepatitis-B do not primarily affect the reproductive tract itself.
(2) Medical Termination of Pregnancy (MTP) is considered relatively safe during the first trimester (upto 12 weeks) of pregnancy.
(3) Non-medicated IUDs are inserted by doctors or expert nurses in the uterus through vagina
(4) Both injections and hormonal IUDs prevent conception by inhibiting ovulation and insemination.

161. Which of the following conditions would lead to disturbance in the genetic equilibrium of a large, isolated population?
(1) Random mating among all individuals
(2) Occurrence of mutations in a gene locus
(3) Complete absence of natural selection pressure
(4) Maintenance of stable gene pool over generations

162. In the evolutionary history of vertebrates, a specific lineage of synapsids gave rise to modern mammals. Choose the most direct ancestor of mammals from the options given below.
(1) Pelycosaurs (2) Therapsids
(3) Thecodonts (4) Sauropsids

163. According to the geological time scale and the evolutionary history of plants, which of the following is the most direct descendant of seed ferns that were abundant during the Jurassic period?
(1) Ginkgos (2) Cycads
(3) Sphenopsids (4) Lycopods

164. Which of the flowing options contributed most significantly to population explosion in India?
(1) Rapid increase in birth rate and maternal mortality rate
(2) Rapid decline in death rate, MMR, and infant mortality rate
(3) Decrease in the number of people in the reproducible age group
(4) Successful implementation of the Reproductive and Child Health care (RCH) programme

165. **Assertion (A):** The probability of a nursing mother conceiving a second child is significantly low during the first six months of intense lactation following parturition.
Reason (R): High levels of prolactin, stimulated by the suckling of the infant, suppress the secretion of gonadotropins from the anterior pituitary.
In the light of the above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

166. Modern *Homo sapiens* arose during the ice age between 75,000-10,000 years ago.
According to the evolutionary timeline, identify the key development that followed the pre-historic cave art.
(1) Burial of the dead
(2) Discovery of fire
(3) Development of agriculture
(4) Use of stone weapons

167. While working on his theory of natural selection, Charles Darwin was significantly influenced by an essay titled on Population. Identify the scholar who wrote this essay and proposed that populations grow geometrically while food sources grow only arithmetically and select the correct option.
(1) Alfred Wallace
(2) Thomas Malthus
(3) Jean-Baptiste Lamarck
(4) Hugo de Vries

168. Louis Pasteur's famous "swan-neck flask" experiment was instrumental in providing evidence for biogenesis. Which of the following correctly identifies the theory he disproved and its main premise?
(1) Theory of Panspermia; Life originated from spores reaching Earth from outer space
(2) Theory of Chemical evolution; Life arose through a series of chemical reactions in a reducing atmosphere
(3) Theory of Spontaneous generation; Life could emerge from decaying and rotting matter like straw or mud
(4) Theory of Special creation; All living organisms were created in their current form by a divine act

Space for Rough Work

169. The development of which adaptation was critical for the definitive transition of vertebrates to fully terrestrial environment, allowing reproduction independent of water?
(1) The formation of a three-chambered heart for efficient circulation
(2) The evolution of moist skin to prevent dehydration
(3) The emergence of the amniotic shelled egg, providing a protective, self-contained aquatic environment for the embryo
(4) The presence of a strong pectoral girdle to support body weight on land

170. The statement that correctly describes the key procedural difference between ZIFT and IUT is that
(1) ZIFT is used only when both fallopian tubes are blocked, while IUT is used for unexplained infertility.
(2) In ZIFT, the zygote or embryo upto 8 blastomeres is transferred into the fallopian tube, whereas in IUT, embryo with more than 8 blastomeres is transferred into the uterus.
(3) ZIFT involves *in-vivo* fertilization followed by embryo transfer, while IUT strictly involves *in- vitro* fertilization.
(4) The success rate of ZIFT is universally higher than IUT due to fertilization occurring within the body.

171. Consider the given statements and choose the **correct** option.
Statement (A): One can reduce the chances of venereal diseases by avoiding sex with unknown partners/multiple partners.
Statement (B): Bacterial STI such as trichomoniasis can be typically cured with an appropriate antibiotic treatment.
(1) Only statement (B) is correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Both statements (A) and (B) are correct

172. Choose the event that took place in the year 1971 among the following.
(1) Government of India legalised MTP with some strict conditions to avoid its misuse
(2) Population of India crossed 1.2 billion
(3) Initiation of programmes called 'family planning' in India 1951
(4) Launching a new oral contraceptive publically for the females which was developed by scientists at Central Drug Research Institute (CDRI) in Lucknow

173. Select the **incorrect** match.
(1) Lichen – Pollution indicator
(2) Conifers – Gave rise to angiosperms
(3) *Brachiosaurus* – Long necked herbivore
(4) *Pteranodon* – Flying reptile

174. Choose the **correct** statement.
(1) *Australopithecus* was the first hominid species known to systematically use fire.
(2) *Dryopithecus* is widely considered the direct ancestor of modern humans based on the current fossil evidence.
(3) Evidence shows that *Australopithecus* hunted with stone weapons but essentially ate fruits.
(4) The Neanderthal man lived in near east and central Africa between 1,00,000-40,000 years back.

175. Identify the contraceptive among the following which is not made up of rubber and select the **correct** option.
(1) Cervical cap
(2) Vault
(3) Condom
(4) Lippes loop

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176. Consider the given diagram representing a method of contraception and select the **incorrect** option w.r.t. it.



(1) Blocks gamete transport from rete testis to vasa efferentia
(2) Highly effective but their reversibility is very poor
(3) Generally advised for the male partner as a terminal method to prevent any more pregnancies
(4) Carried out by making a small incision on the scrotum

177. Read the given statements and choose the **correct** option.

Statement (A): Among the stories of evolution of individual species, the story of evolution of modern man is most interesting and appears to parallel evolution of human brain and language.

Statement (B): We can call human evolution as the classic case of adaptive radiation.

(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct

178. A large, diverse population of a flightless bird is drastically reduced to only a few individuals after a volcanic eruption. This genetic makeup of the surviving population is an example of which evolutionary force?

(1) Gene flow (2) Directional selection
(3) Founder effect (4) Bottleneck effect

179. Comprehend the given statements.

(a) Progestasert as well as CuT prevent conception by inhibiting ovulation and oogenesis.
(b) Contraceptives are regular requirements for the maintenance of reproductive health.
(c) Contraceptive pills create a sterile inflammatory response in the vagina unsuitable for implantation.

Select the **correct** option.

(1) Both (a) and (b) are true; (c) is false
(2) (a), (b) and (c) are false
(3) Both (a) and (b) are false; (c) is true
(4) Only (b) and (c) are true

180. Choose the **incorrect** statement.

(1) The evolutionary force that leads to founder effect in a population is genetic drift
(2) According to Darwin, adaptive ability is inherited
(3) The mechanism of genetic variation in the population of a sexually reproducing organism is genetic recombination
(4) According to Lamarck, the organic evolution is due to descent with modification of organisms

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